PAD: Phase–Amplitude Decoupling Fusion for Multimodal Land Cover Classification

Huiling Zheng, Xian Zhong, *Senior Member, IEEE*, Bin Liu, *Member, IEEE*, Yi Xiao, *Member, IEEE*, Bihan Wen, *Senior Member, IEEE*, and Xiaofeng Li, *Fellow, IEEE*

***Abstract*—The fusion of synthetic aperture radar (SAR) and RGB imagery for land cover classification (LCC) remains challenging due to modality heterogeneity and underexploited spectral complementarity. Existing approaches often fail to decouple shared structural features from modality-complementary radiometric attributes, resulting in feature conflicts and information loss. To address this, we propose phase–amplitude decoupling (PAD), a frequency-aware framework that separates phase (modality-shared) and amplitude (modality-complementary) components in the Fourier domain. This design reinforces shared structures while preserving complementary characteristics, thereby enhancing fusion quality. Unlike previous methods that overlook the distinct physical properties encoded in frequency spectra, PAD explicitly introduces amplitude–phase decoupling for multimodal fusion. Specifically, PAD comprises two key components: 1) phase spectrum correction (PSC), which aligns cross-modal phase features via convolution-guided scaling to improve geometric consistency; and 2) amplitude spectrum fusion (ASF), which dynamically integrates high-frequency (HF) and low-frequency (LF) patterns using frequency-adaptive multilayer perceptrons (MLP), effectively exploiting SAR's morphological sensitivity and RGB's spectral richness. Extensive experiments on WHU-OPT-SAR and DDHR-SK demonstrate state-of-the-art performance. This work establishes a new paradigm for physics-aware multimodal fusion in remote sensing. The code will be available at https://github.com/RanFeng2/PAD**

Received 3 July 2025; revised 27 September 2025; accepted 7 October 2025. Date of publication 15 October 2025; date of current version 31 October 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 62271361 and Grant 42006159 and in part by Hubei Provincial Key Research and Development Program under Grant 2024BAB039. *(Corresponding authors: Xian Zhong; Bin Liu.)*

Huiling Zheng is with Sanya Science and Education Innovation Park, Wuhan University of Technology, Sanya 572025, China, also with the School of Computer Science and Artificial Intelligence, Wuhan University of Technology, Wuhan 430070, China, and also with the Key Laboratory of Ocean Circulation and Waves, Institute of Oceanology, Chinese Academy of Sciences, Qingdao 266071, China (e-mail: zhenghl@whut.edu.cn).

Xian Zhong is with Hubei Key Laboratory of Transportation Internet of Things, School of Computer Science and Artificial Intelligence, Wuhan University of Technology, Wuhan 430070, China, and also with the State Key Laboratory of Maritime Technology and Safety, Wuhan University of Technology, Wuhan 430063, China (e-mail: zhongx@whut.edu.cn).

Bin Liu is with the College of Oceanography and Ecological Science, Shanghai Ocean University, Shanghai 201306, China (e-mail: bliu@shou.edu.cn).

Yi Xiao is with the School of Computer and Artificial Intelligence, Zhengzhou University, Zhengzhou 450001, China (e-mail: yixiao@zzu.edu.cn).

Bihan Wen is with the Rapid-Rich Object Search Laboratory, School of Electrical and Electronic Engineering, Nanyang Technological University, Singapore 639798 (e-mail: bihan.wen@ntu.edu.sg).

Xiaofeng Li is with the Key Laboratory of Ocean Circulation and Waves, Institute of Oceanology, Chinese Academy of Sciences, Qingdao 266071, China (e-mail: xiaofeng.li@ieee.org).

Digital Object Identifier 10.1109/TGRS.2025.3621902

***Index Terms*—Land cover classification (LCC), multimodal segmentation, remote sensing, RGB–synthetic aperture radar (SAR) multimodality.**

I. Introduction

LAND cover classification (LCC) is a fundamental Earth observation task that categorizes surface materials, such as farmlands, forests, water bodies, and urban areas, using remote sensing imagery. This capability is essential for environmental monitoring, resource management, and urban development [1]. Recently, fusing synthetic aperture radar (SAR) and RGB imagery has emerged as a promising approach to enhance LCC performance: SAR provides robust structural information resilient to atmospheric and illumination variations, while RGB imagery offers rich spectral details that mitigate SAR artifacts, such as speckle noise [2], [3].

Despite this complementarity, RGB–SAR fusion faces a key challenge: the inherent heterogeneity between microwave backscattering (SAR) and optical reflectance (RGB) causes spectral misalignment and hinders effective cross-modal feature integration [see Fig. 1(a)]. In SAR images, objects often appear as discrete backscatter responses, lacking fundamental visual cues, such as geometric contours, textures, and consistent shapes, that are readily available in RGB imagery [4]. Compared with natural image fusion, integrating SAR and RGB data presents pronounced radiometric and geometric disparities, which limit existing frameworks [5], [6], [7].

Deep learning, particularly convolutional neural networks (CNNs), has achieved remarkable success in LCC by extracting powerful features [8], [9], [10], [11]. To mitigate modality heterogeneity, most approaches employ dual-branch architectures to extract multiscale features independently and then fuse them via element-wise summation, mean aggregation, or channel-wise concatenation before a shared decoder or segmentation head [12]. However, accurately modeling complementary features remains impeded by SAR–RGB heterogeneity. Inspired by contrastive learning [13], [14], recent works align modality-shared features to guide complementary feature learning and alleviate intermodal conflicts [see Fig. 1(b)], yet spatial-domain modeling remains inefficient.

To better capture RGB–SAR complementarity, some methods explore frequency-domain fusion [15], [16], [17]. For example, Li et al. [18] treat phase as modality-shared features but discard amplitude information essential for terrain delineation and do not systematically validate phase sharing.

We conduct a comprehensive spectral analysis [see Fig. 1(c)–(g)] to reveal three critical frequency-domain priors:

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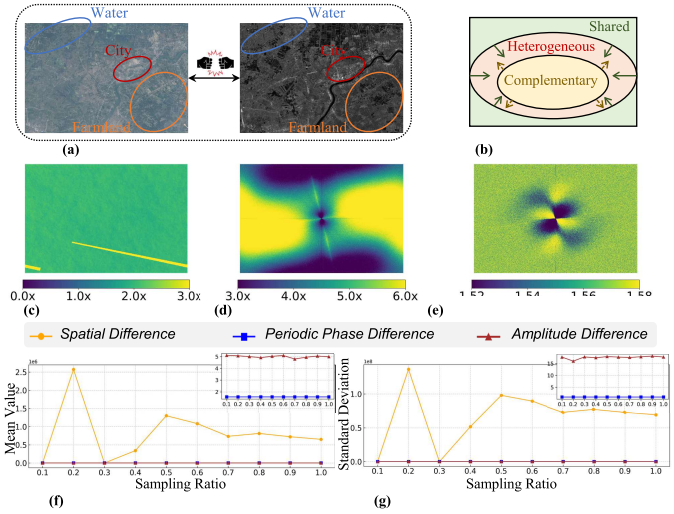


Fig. 1. Motivation and SAR–RGB spatial–frequency difference analysis. (a) SAR and RGB imagery exhibit modality gaps due to distinct sensing mechanisms (backscattering versus reflectance). (b) Fusion feature space contains heterogeneous, shared, and complementary components; effective fusion suppresses heterogeneous cues while enhancing shared and complementary patterns. Frequency-domain analysis on WHU-OPT-SAR (see Section III-A), (c) spatial discrepancies show anisotropic distributions, (d) relative amplitude differences concentrate in high-frequency (HF) regions, (e) phase discrepancies are minimal and primarily cluster in low-frequency (LF) bands, and (f) and (g) varying the sampling rate significantly affects spatial discrepancy distributions.

1) SAR–RGB spatial-domain differences are anisotropic and susceptible to pixel-level mismatch [see Fig. 1(c)]; 2) average phase spectra differ minimally and primarily in low frequencies, indicating phase as shared information [see Fig. 1(e)]; and 3) amplitude mismatches exhibit distinct HF and LF patterns, most pronounced in the HF band, highlighting the need for frequency-aware modeling [see Fig. 1(d)].

Motivated by these insights, we propose phase–amplitude decoupling (PAD), which decouples phase components capturing modality-shared structures from amplitude components encoding modality-complementary details. By explicitly aligning shared phase representations, PAD enhances complementary amplitude feature extraction and reduces cross-modal conflicts. PAD comprises two modules: phase spectrum correction (PSC), which aligns and reinforces structural details via convolution-guided scaling; and amplitude spectrum fusion (ASF), which dynamically integrates HF and LF patterns using frequency-adaptive multilayer perceptrons (MLPs).

Our contributions are fourfold.

1) We systematically analyze registered datasets to establish the frequency distributions of SAR–RGB shared (phase) and complementary (amplitude) patterns, providing interpretable physical priors for multisource fusion.
2) We introduce frequency-domain amplitude–phase decoupling for multimodal fusion, orthogonally separating phase (modality-shared) and amplitude (modality-complementary) features to resolve heterogeneity-induced conflicts.
3) PSC employs convolution-guided multiplicative scaling to align and reinforce critical structural details for robust fusion.
4) ASF leverages frequency-adaptive MLPs to integrate complementary HF (local details) and LF (global structure) components, exploiting SAR's morphological sensitivity and RGB's spectral richness for improved segmentation.

II. Related Work

A. Multimodal Fusion for LCC

Single-modality approaches [19], [20], [21], [22], [23], [24], [25], [26] perform well on homogeneous data but cannot exploit cross-modal synergies in complex environments. SAR–RGB fusion methods generally follow two paradigms: cross-modal transfer learning and hierarchical feature integration.

1) Cross-Modal Transfer Learning: Transfer learning [27] adapts knowledge from one modality to another, but the pronounced differences between SAR and RGB often degrade learned representations, limiting their multimodal applicability.

2) Hierarchical Feature Fusion: Feature fusion has evolved through three stages.

a) *Basic Fusion Operators:* Early methods employed concatenation or weighted summation [28], but these are prone to modality dominance and redundancy.
b) *Attention-Guided Fusion:* Spatial–channel attention mechanisms alleviate these issues [29], [30]. For example, MCANet's cross-modal attention (MCAM) models intermodality dependencies, MRFS [31] introduces progressive cycle attention for infrared–visible fusion; and CMX [32] applies bidirectional channel–spatial attention. However, these paradigms incur high computational costs.
c) *Complementarity-Aware Fusion:* Recent approaches explicitly decouple shared and complementary features. ASANet [33] uses semantic focusing modules to emphasize complementary cues, CDDFuse [13] applies contrastive learning to separate private and shared representations, and CEN [34] leverages dynamic channel exchange with independent batch normalization. Despite these advances, spatial-domain methods still struggle with geometric misalignment and inefficiency.

B. Frequency Representation Learning in Vision Tasks

Frequency representation learning has emerged as a powerful tool across various vision tasks due to its capacity for global modeling and structural decomposition. We briefly review three relevant areas.

1) Spectral Decomposition for Structure-Style Separation: Fourier-based decomposition isolates phase (structural semantics) and amplitude (appearance) for domain generalization and style transfer [35]. Early methods relied on feature statistics for style normalization [36], [37], while recent works apply Fourier transforms for clearer separation, enabling content-preserving augmentation by swapping amplitudes across domains [38], [39].

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2) Frequency-Driven Global Modeling: Fourier transforms provide a global receptive field useful for efficient attention and context modeling. Fast Fourier convolution (FFC) [40] and FFTformer [41] leverage spectral processing to replace or accelerate spatial attention. Spectral token mixing has also proven effective in lightweight transformers [42], [43].

3) Frequency-Based Inductive Biases in Dense Prediction: Spectral priors benefit dense prediction tasks [44], [45], [46], [47]. DFF [48] uses frequency-domain dynamic filtering to amplify invariant LF features, improving cross-domain generalization. FreqFusion [49] designs learnable high- and low-pass filters for semantic consistency and boundary refinement. FCENet [15] employs frequency-selective modules to model cross-modal spectral correlations in RGB–NIR denoising.

In summary, frequency-based representations offer a flexible and physically interpretable design space for vision tasks. Yet, few frameworks address task-specific fusion in remote sensing. We target this gap in Section III.

C. Frequency-Domain Fusion Methods

The application of frequency representations to multimodal fusion is relatively recent. Shi et al. [16] introduced Frefusion, which applies spectral attention to merge modalities in the frequency domain. Frequency spectra naturally decompose signals into phase (modality-shared structures) and amplitude (modality-complementary details), offering a principled basis for fusion.

However, existing methods often transplant spatial operators, such as MLPs [17], convolutions [15], or attention [16], into the spectral domain, modulating magnitude while neglecting the distinct roles of phase and amplitude. As a result, shared information may be preserved, but complementary cues are frequently suppressed.

In contrast, our PAD framework explicitly decouples phase and amplitude spectra. Through systematic spectral analysis (see Section III-A), we show that SAR and RGB exhibit strong phase consistency, while amplitude captures complementary patterns. PAD introduces: 1) a PSC module, which performs robust structural alignment and 2) an ASF module, which conducts frequency-adaptive refinement. This decomposition-correction–fusion pipeline distinguishes PAD from previous FFT-based approaches by providing a physically interpretable strategy for multimodal fusion.

The two most related works, FeINFN [17] and SFINet++ [50], also leverage amplitude and phase in the frequency domain for HF detail enhancement and global context modeling. However, they lack a physically grounded interpretation of amplitude–phase roles and do not explicitly decouple shared versus complementary information. In contrast, PAD is the first framework to formulate and exploit amplitude–phase decomposition for principled alignment and complementary integration across modalities.

III. Proposed Method

To address the fundamental conflict between modality-shared geometry and modality-complementary appearance in SAR–RGB fusion, we introduce PAD, a frequency-aware multimodal fusion framework. The overall architecture in Fig. 2 comprises three modules: the spatial complementary fusion (SCF) module, which preserves fine-grained spatial priors; the PSC module, which aligns and reinforces shared structural details; and the ASF module, which adaptively integrates complementary HF and LF amplitude cues. Unlike previous methods that apply spectral operations uniformly, PAD explicitly decouples and then recombines phase and amplitude with dedicated modules for physics-consistent fusion.

Algorithm 1 Spectral Difference Analysis

Input : SAR/RGB image pairs $\{I_{\text{SAR}}^{(i)}, I_{\text{RGB}}^{(i)}\}_{i=1}^{N}$
Output: RSD, RAD, APPD metrics for each image pair

1: Preprocessing
//Convert RGB image to grayscale
$I_{\text{RGB}} \leftarrow 0.299R + 0.587G + 0.114B$ //Retain SAR image as is
$I_{\text{SAR}} \leftarrow I_{\text{SAR}}$
2: Spectral Decomposition
for *each pair* (i) **do**
  //Frequency domain conversion
  Compute $\mathcal{A}_{\text{SAR}}^{(i)}$ and $\mathcal{P}_{\text{SAR}}^{(i)}$ via 2D FFT and FFTshift; Compute $\mathcal{A}_{\text{RGB}}^{(i)}$ and $\mathcal{P}_{\text{RGB}}^{(i)}$ similarly
  //Spatial difference computation
  Compute $\Delta I^{(i)} \leftarrow \text{RSD}^{(i)}$ via Eq. (1) //Amplitude difference computation
  Compute $\Delta \mathcal{A}^{(i)} \leftarrow \text{RAD}^{(i)}$ via Eq. (2) //Phase difference computation
  Compute $\Delta \mathcal{P}^{(i)} \leftarrow \text{APPD}^{(i)}$ via Eq. (3)
end
3: Dataset-level Aggregation
//Mean spatial, amplitude, phase difference computation
RSD $\leftarrow$ Mean$(\{\Delta I^{(i)}\}_{i=1}^{N})$ RAD $\leftarrow$ Mean$(\{\Delta \mathcal{A}^{(i)}\}_{i=1}^{N})$ APPD $\leftarrow$ Mean$(\{\Delta \mathcal{P}^{(i)}\}_{i=1}^{N})$

A. Frequency-Domain Priors for Amplitude–Phase Decoupling

To validate the physical significance of amplitude–phase decoupling and reveal the shared and complementary relations between RGB and SAR, we conduct a systematic spectral analysis on WHU-OPT-SAR, the largest RGB–SAR segmentation benchmark.

First, we visualize averaged *difference spectra* [see Fig. 1(c)–(e)] using Algorithm 1 to compare modality differences in the spatial, amplitude, and phase domains. This analysis shows that frequency-domain representations capture cross-modal sharing and complementarity more effectively than spatial representations.

The difference spectra are defined as

$$\text{RSD} = \frac{1}{N}\sum_{i=1}^{N}\frac{|I_{\text{SAR}}^{(i)} - I_{\text{RGB}}^{(i)}|}{I_{\text{RGB}}^{(i)} + \epsilon} \tag{1}$$

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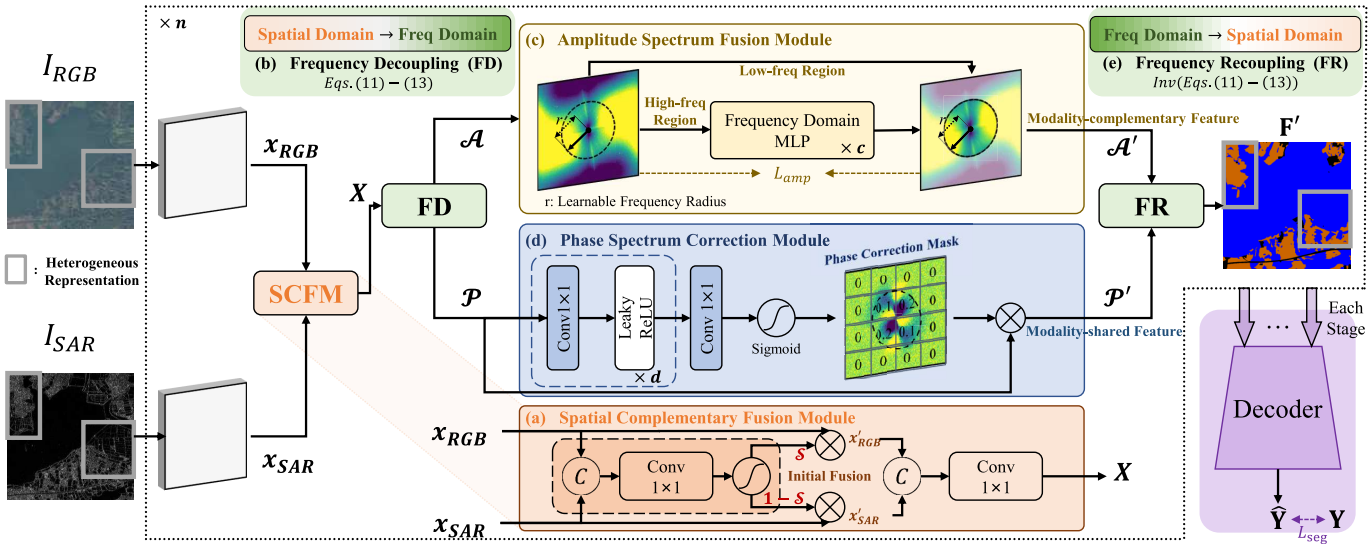


Fig. 2. Flowchart of the PAD framework. Registered RGB and SAR inputs, I_{RGB} and I_{SAR}, are processed by asymmetric backbones to extract features x_{RGB} and x_{SAR} at each stage. These features pass through n PAD fusion modules, each comprising (a) SCF, (b) FD (spatial-to-frequency transform), (c) ASF, (d) PSC, and (e) FR (frequency-to-spatial transform). At each stage, the fused features are concatenated channel-wise (outer circle, C) and fed into a shared decoder. Here, $\mathcal{A}$ and $\mathcal{P}$ denote the amplitude and phase spectra, respectively.

$$\mathrm{RAD} = \frac{1}{N}\sum_{i=1}^{N}\frac{\left|\mathcal{A}_{\mathrm{SAR}}^{(i)} - \mathcal{A}_{\mathrm{RGB}}^{(i)}\right|}{\mathcal{A}_{\mathrm{RGB}}^{(i)} + \epsilon} \tag{2}$$

$$\mathrm{APPD} = \frac{1}{N}\sum_{i=1}^{N}\left|\arg\exp\left(i\left(\mathcal{P}_{\mathrm{SAR}}^{(i)} - \mathcal{P}_{\mathrm{RGB}}^{(i)}\right)\right)\right| \tag{3}$$

where I denotes pixel intensity; $\mathcal{A}$ and $\mathcal{P}$ are amplitude and phase spectra, respectively; $\arg(\cdot)$ returns the complex angle; $i = \sqrt{-1}$; and $\epsilon > 0$ prevents division by zero. By leveraging the periodicity of complex exponentials, (3) maps phase differences to $[-\pi, \pi)$, correcting wrap-around artifacts (e.g., $2\pi \to 0$).

Fig. 1(c)–(e) reveals three priors: 1) spatial differences (RSD) exhibit anisotropic distributions; 2) amplitude differences (RAD) concentrate in HF regions; and 3) phase differences [average periodic phase difference (APPD)] remain minimal, with slight LF clustering. These frequency-domain metrics provide better explainability than spatial RSD, confirming that spectral decomposition more faithfully captures cross-modal characteristics.

Physically, LF phase disparities stem from geometric distortions (e.g., side-looking SAR versus orthographic RGB), while HF variations arise from SAR speckle. Amplitude disparities reflect modality-complementary sensitivity to object scales: low frequencies encode large structures (farmlands and forests), whereas high frequencies capture fine details (roads).

These analyses yield two key insights that guide our design: 1) phase sharing: consistently low APPD indicates strong phase consistency; PSC, therefore, performs pixel-level alignment while preserving this shared information; and 2) amplitude complementarity: the frequency-dependent divergence of RAD motivates ASF's dynamic frequency–radius adjustment to handle amplitude variations.

Second, we conduct sensitivity experiments on sampling rates [see Fig. 1(f) and (g)]. By downsampling images at various ratios and computing the mean/variance of RSD, RAD, and APPD across the dataset, we observe that RAD and APPD remain nearly constant, whereas RSD fluctuates more. This invariance indicates that frequency-domain complementarity arises from intrinsic modality characteristics rather than sampling artifacts.

TABLE I
STATISTICAL ANALYSIS OF THE APPD SPECTRUM

Metric	Low frequency (LF)	High frequency (HF)	ALL
Shapiro-Wilk statistic	1.0044	1.0050	1.0131
Shapiro-Wilk p-value	1.0000	1.0000	1.0000
Skewness	-0.0521	0.0020	-0.0265
Kurtosis	0.0750	-0.0117	0.0322

Third, we assess statistical consistency of cross-modal phase spectra via Shapiro–Wilk tests and by computing skewness and kurtosis of APPD within each frequency band (see Table I). Frequency histograms and mean–variance trajectories (see Fig. 3) show that APPD distributions closely follow a Gaussian profile, especially in HF regions; small LF deviations likely result from structural noise. In summary, RGB and SAR phases are highly similar and largely unaffected by instance-level variations.

Finally, we evaluate phase invariance under two challenging scenarios: 1) partial RGB removal (see Fig. 4) and 2) cloud occlusion (see Fig. 5). In the former, RSD increases markedly when RGB pixels are missing [see Fig. 4(d) and (g)], whereas APPD remains stable [see Fig. 4(f) and (i)]. In the latter, the variance of APPD ($\Delta \approx 10^{-0.037}$) is negligible compared with that of RSD ($\Delta \approx 10^{5.81}$) and RAD ($\Delta \approx 10^{0.58}$) in Fig. 5(d). These results confirm that phase, particularly at high frequencies, retains robust cross-modal consistency, underscoring its role as a shared feature to mitigate intermodal conflicts and enhance modality-complementary fusion.

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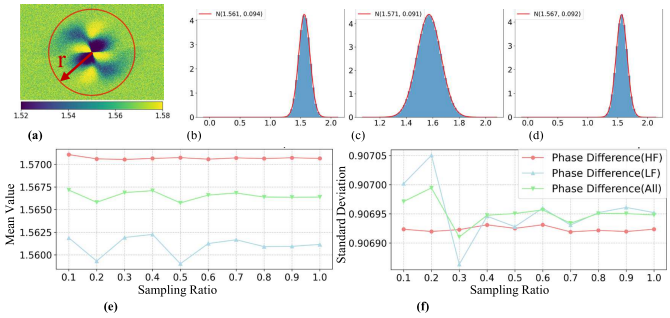


Fig. 3. Statistical Analysis of APPD. (a) APPD spectrum. (b)–(d) Frequency histograms with Gaussian fits for LF; (radius = 0.5× half-diagonal), HF, and all spectra (ALL). (e) and (f) Mean–variance trajectories across sampling rates. Across sampling rates, ALL exhibit near-Gaussian behavior, while slight deviations in LF and ALL are attributable to structural noise at low frequencies, indicating that the phase, particularly HF, serves as a robust shared feature.

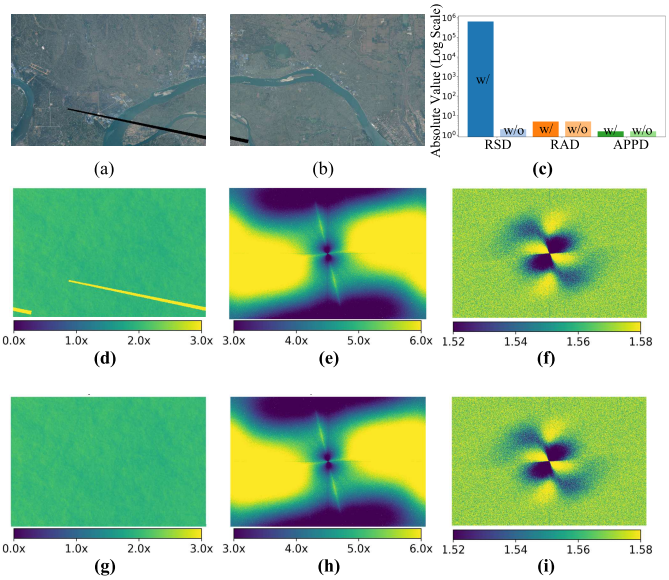


Fig. 4. Visualization of spatial–frequency spectrum differences under missing RGB regions. (a) and (b) Two RGB images have missing regions in the original dataset. (c) Variation in RSD, RAD, and APPD, indicating RSD's greater sensitivity to missing data. (d)–(f) Difference spectra computed, *including* the missing regions. (g)–(i) Difference spectra computed, *excluding* the missing regions. Overall, frequency-domain metrics (RAD and APPD) are more robust to missing data than the spatial–domain RSD.

B. Overall Framework

As illustrated in Fig. 2, PAD integrates SAR and RGB via a dual-branch encoder and a shared decoder, with its primary innovation lying in cross-modal feature fusion through PAD. The encoder employs *asymmetric convolutional backbones* $\mathcal{B}_{\text{SAR}}$ and $\mathcal{B}_{\text{RGB}}$, each with independent parameters to preserve modality-complementary attributes. Specifically, $\mathcal{B}_{\text{SAR}}$ extracts microwave backscattering features from SAR images, while $\mathcal{B}_{\text{RGB}}$ captures optical reflectance patterns from RGB images. Both backbones produce hierarchical features at four stages with resolutions $\{1/4, 1/8, 1/16, 1/32\}$ for $i \in \{0, 1, 2, 3\}$

$$x^i_{\text{RGB}} = \mathcal{B}^{(i)}_{\text{RGB}}\left(I_{\text{RGB}}\right) \quad x^i_{\text{SAR}} = \mathcal{B}^{(i)}_{\text{SAR}}\left(I_{\text{SAR}}\right). \tag{4}$$

Although frequency-domain fusion is robust to modality differences, it may discard rich local interactions. To retain

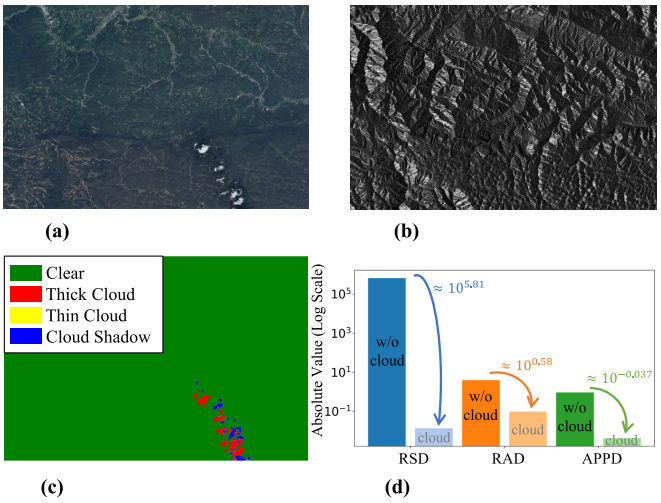


Fig. 5. Visualization of spatial–frequency spectrum differences under cloud obscuration. (a)–(c) Example observations with cloud masks derived from the RGB, SAR, and NIR channels in WHU-OPT-SAR [51], respectively. (d) Comparison of RSD, RAD, and APPD between cloud-covered and cloud-free areas; arrows indicate the value of the decrease.

fine-grained spatial priors, we introduce a lightweight SCF module. SCF computes pixel-wise fusion weights via spatial attention gating and reduces feature dimensionality to constrain the solution space for subsequent spectral operations

$$X^i = \text{SCF}\left(x^i_{\text{RGB}}, x^i_{\text{SAR}}\right) \tag{5a}$$

$$\mathcal{S} = \sigma\left(\text{Conv}_{1\times1}\left(\text{Concat}\left(x_1, x_2\right)\right)\right) \tag{5b}$$

$$x'_1 = \mathcal{S} \odot x_1 \quad x'_2 = (1 - \mathcal{S}) \odot x_2 \tag{5c}$$

$$x = \text{Conv}_{1\times1}\left(\text{Concat}\left(x'_1, x'_2\right)\right) \tag{5d}$$

where $(x_1, x_2) = (x^i_{\text{RGB}}, x^i_{\text{SAR}})$, $\sigma(\cdot)$ is the Sigmoid, $\text{Conv}_{1\times1}$ denotes a 1×1 convolution, $\text{Concat}(\cdot,\cdot)$ concatenates along channels, and $\odot$ denotes the Hadamard (element-wise) product.

The fused feature X^i is then transformed to separate amplitude $\mathcal{A}^i$ and phase $\mathcal{P}^i$ spectra

$$\mathcal{A}^i, \mathcal{P}^i = \text{FD}\left(X^i\right) \tag{6}$$

where $\text{FD}(\cdot)$ denotes the frequency decoupling (FD) operator (defined in Section III-C). ASF enhances the amplitude spectrum using a trainable frequency radius r to distinguish and reinforce HF details while preserving LF structures

$$\mathcal{A}'^i = \text{ASF}\left(\mathcal{A}^i; r\right). \tag{7}$$

PSC applies multiplicative scaling to correct the phase

$$\mathcal{P}'^i = \text{PSC}\left(\mathcal{P}^i\right). \tag{8}$$

Finally, inverse reconstruction produces fused features

$$\boldsymbol{F}'^i = \text{FR}\left(\mathcal{A}'^i, \mathcal{P}'^i\right) \tag{9}$$

where $\text{FR}(\cdot,\cdot)$ is the inverse frequency recoupling operator (defined in Section III-C). A shared decoder $\mathcal{D}$ aggregates multistage fused features to generate the land-cover map

$$\widehat{\boldsymbol{Y}} = \mathcal{D}\left(\boldsymbol{F}'^0, \boldsymbol{F}'^1, \boldsymbol{F}'^2, \boldsymbol{F}'^3\right). \tag{10}$$

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TABLE II

QUANTITATIVE RESULTS (%) ON WHU-OPT-SAR. PER-CLASS IOU IS REPORTED FOR SEVEN LAND-COVER CATEGORIES (FARMLAND, CITY, VILLAGE, WATER, FOREST, ROAD, AND OTHERS), ALONG WITH OA, MKAPPA, MF1, AND MIOU. THE BEST AND SECOND-BEST RESULTS ARE HIGHLIGHTED IN **BOLD** AND UNDERLINED, RESPECTIVELY

	Method	Venue	Per-class IoU							OA	mKappa	mF1	mIoU
			Farmland	City	Village	Water	Forest	Road	Others				
RGB	DANet [19]	CVPR'19	67.51	55.22	46.71	60.85	83.13	38.89	20.43	83.19	49.74	67.40	53.25
	CGNet [21]	TIP'20	61.33	45.96	37.98	49.33	80.98	1.92	0.19	79.26	37.42	50.54	39.67
	OCRNet [20]	ECCV'20	67.68	56.48	47.47	61.22	82.98	37.20	17.97	83.23	49.11	66.95	53.00
	SETR [23]	CVPR'21	65.25	55.07	42.97	57.80	82.44	31.91	18.36	81.94	46.19	64.73	50.54
	SegFormer [22]	NeurIPS'21	68.18	56.91	46.82	60.85	83.19	35.24	18.73	83.37	49.46	66.79	52.84
	Mask2Former [24]	CVPR'22	67.09	56.64	43.97	60.09	81.47	36.24	21.03	82.69	47.69	66.65	52.36
	PIDNet [25]	CVPR'23	66.20	55.20	44.95	58.90	82.20	36.36	16.40	82.25	47.20	65.53	51.46
	TransUNet [26]	MIA'24	67.62	56.28	46.08	60.66	83.25	38.76	19.32	83.18	49.89	67.20	53.14
SAR	DANet [19]	CVPR'19	60.42	54.12	33.05	54.45	80.57	24.89	15.11	79.44	35.28	60.16	46.09
	CGNet [21]	TIP'20	56.72	46.81	26.32	46.30	77.53	0.27	0.00	76.39	24.64	47.00	36.28
	OCRNet [20]	ECCV'20	59.81	52.48	31.21	54.03	79.57	22.49	13.92	78.94	32.62	58.74	44.79
	SETR [23]	CVPR'21	60.35	52.29	31.41	53.56	80.15	18.66	15.18	79.12	34.25	58.33	44.51
	SegFormer [22]	NeurIPS'21	60.17	54.71	32.30	54.07	79.79	21.02	15.13	79.31	33.60	59.24	45.31
	Mask2Former [24]	CVPR'22	60.39	52.86	35.69	53.85	78.36	29.16	17.20	79.26	32.76	61.35	46.79
	PIDNet [25]	CVPR'23	58.59	52.35	29.65	53.60	79.35	23.06	13.74	78.42	32.15	58.33	44.34
	TransUNet [26]	MIA'24	61.93	54.85	33.42	56.26	80.63	27.56	14.96	80.14	37.45	61.13	47.09
SAR & RGB	MCANet [29]	JAG'21	65.14	53.97	45.43	54.59	81.87	31.37	13.43	81.74	44.28	63.37	49.40
	DDHRNet [30]	JAG'22	64.80	49.42	40.98	54.11	81.31	22.96	10.32	80.94	44.49	59.84	46.27
	CEN [34]	TPAMI'22	67.60	55.73	46.46	60.65	83.28	34.22	15.94	83.17	49.37	65.79	51.98
	TokenFusion [55]	CVPR'22	68.43	58.34	45.76	61.09	83.79	32.03	15.20	83.65	49.45	65.67	52.09
	CDDFuse [13]	CVPR'23	60.07	48.64	23.96	50.67	79.47	15.16	11.43	78.02	51.36	54.54	41.34
	CMX [32]	TITS'23	**70.04**	**58.41**	49.04	64.77	84.12	41.76	22.21	84.53	53.46	69.60	55.76
	FTransUNet [6]	TGRS'24	68.22	56.28	46.76	62.35	83.46	35.35	17.23	83.50	50.94	66.61	52.81
	ASANet [33]	ISPRS'24	69.71	58.09	49.05	63.52	84.05	41.50	20.57	84.29	52.83	69.04	55.21
	MRFS [31]	CVPR'24	69.59	58.04	49.15	63.78	**84.15**	41.26	22.88	84.24	**54.14**	69.48	55.55
	ASMFNet [56]	JSTARS'24	67.31	55.73	46.56	59.51	83.03	32.63	14.60	82.91	49.51	65.09	51.34
	MIEFNet [57]	GRSL'24	68.72	57.16	48.04	63.23	83.64	39.35	18.41	83.87	50.58	67.89	54.08
	MFNet [58]	TGRS'25	68.97	58.39	47.78	63.38	83.88	39.51	20.33	84.02	51.86	68.47	54.61
	PAD (Ours)		69.85	58.10	**49.70**	**65.39**	84.05	**43.24**	**23.47**	**84.56**	53.93	**70.14**	**56.26**

C. FD and Recoupling

To obtain physically decoupled spectral representations, we employ a bidirectional transformation comprising FD and inverse recoupling (FR). Given $X \in \mathbb{R}^{H\times W}$, a 2-D real-input fast Fourier transform (rFFT) [52] yields

$$\mathcal{F}(u,v) = \sum_{y=0}^{H-1}\sum_{x=0}^{W-1} X(y,x)\exp\left(-\mathrm{i}2\pi\left(\frac{ux}{W}+\frac{vy}{H}\right)\right) \tag{11}$$

where (u, v) are frequency coordinates. Exploiting conjugate symmetry for real inputs, we retain only the nonredundant region and center LF components via an FFT shift

$$\mathcal{F}_{\text{shift}}(u,v) = \mathcal{F}\left(u\oplus\frac{W}{2}, v\oplus\frac{H}{2}\right) \tag{12}$$

where $\oplus$ is modulo addition. Amplitude and phase are

$$\mathcal{A} = |\mathcal{F}_{\text{shift}}| \quad \mathcal{P} = \angle\mathcal{F}_{\text{shift}} \tag{13}$$

with $\mathcal{A}, \mathcal{P} \in \mathbb{R}^{H\times(\lfloor W/2\rfloor+1)}$. FR applies the inverse operations to reconstruct spatial features from $(\mathcal{A}, \mathcal{P})$.

D. PSC Module

While prior works often assume perfect phase alignment [18], our spectral analysis [see Fig. 1(e)] shows that LF geometric inconsistencies persist, necessitating a learnable, spatially aware correction mechanism. Inspired by channel attention, we propose a lightweight PSC module tailored for frequency-domain phase alignment.

As shown in Fig. 1(e), minor misalignments in the LF phase spectra between SAR and RGB, attributable to geometric distortions (e.g., side-looking SAR versus orthographic RGB), can compromise spatial consistency and boundary precision. Traditional rigid registration is inadequate for such nonrigid distortions. PSC performs adaptive geometric correction using learnable multiplicative masks.

Unlike global channel scaling in traditional attention, PSC estimates a spatially varying modulation mask via a stack of d consecutive 1×1 convolutions, enabling localized, fine-grained phase adjustment. Given an input phase–spectrum tensor $P \in \mathbb{R}^{C\times H\times(\lfloor W/2\rfloor+1)}$, we first compress channels with d successive 1×1 convolutions (reducing C to C/r, $r \in \{4, 8\}$), each followed by a Leaky ReLU [53]. A final 1×1 convolution produces a single-channel phase correction mask (PCM), constrained to $(0, 1)$ by a Sigmoid

$$\text{PCM} = \sigma\left(\text{Conv}_d^{1\times1}\left(\delta\left(\cdots\delta\left(\text{Conv}_1^{1\times1}(P)\right)\cdots\right)\right)\right) \tag{14}$$

where $\delta(\cdot)$ is Leaky ReLU and $\text{Conv}_k^{1\times1}$ denotes the kth 1×1 convolution in the stack. Pixel-wise multiplicative scaling is then applied

$$P' = P\odot(1+\text{PCM}). \tag{15}$$

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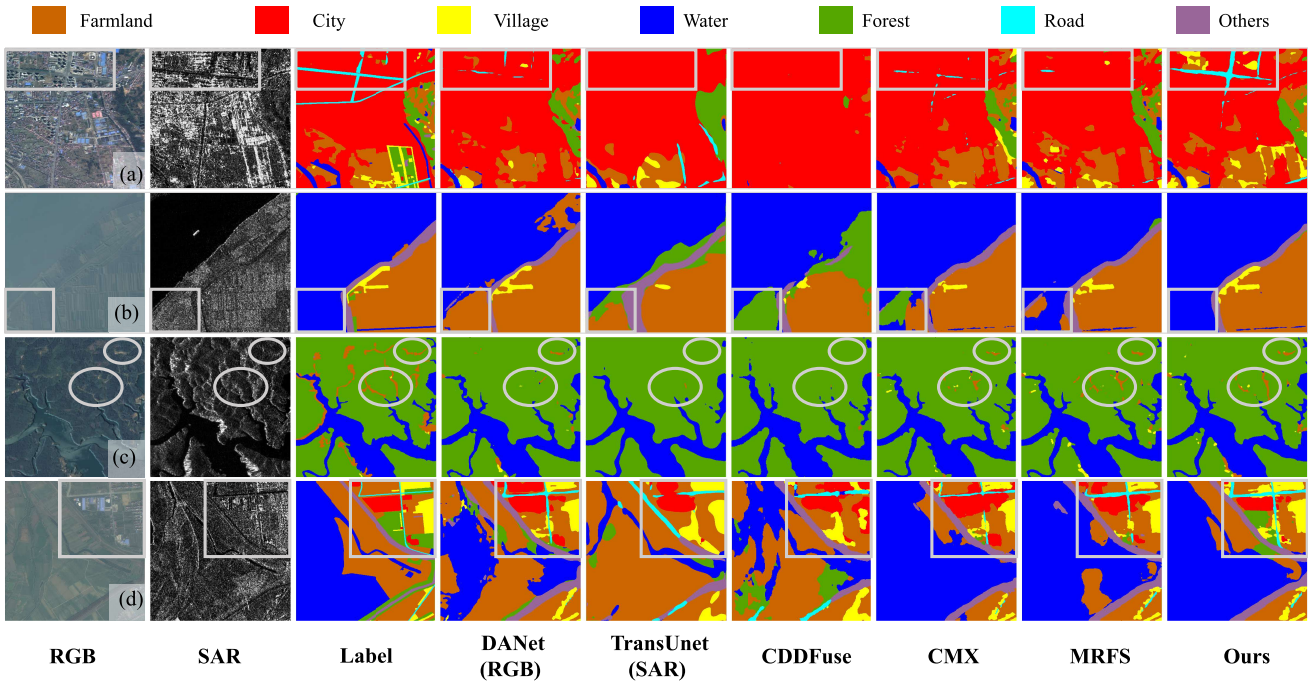


Fig. 6. Visualization results on WHU-OPT-SAR. Four representative scenarios (a) building-dense area, (b) specular water bodies, (c) vegetated region, and (d) urban–rural transition. For each scenario, images are shown (left to right) RGB, SAR, pseudo-color ground truth, and predictions from six competing methods. We compare PAD against unimodal baselines (DANet-RGB and TransUNet-SAR) and multimodal approaches (CDDFuse, CMX, and MRFS). Regions of interest are highlighted with gray boxes (buildings/water) and circles (vegetation); see Section IV-B for details.

This formulation enables localized, fine-grained phase adjustment while preserving phase periodicity and incurring minimal computational overhead.

E. ASF Module

Previous frequency-domain methods often employ fixed high/low-pass filters or spatially uniform spectral attention, which cannot adapt to content-specific frequency distributions. In contrast, our analysis in Fig. 1(d) shows that RGB–SAR spectral discrepancies are concentrated in medium-to-high amplitude frequencies. To exploit this prior while maintaining gradient stability, ASF uses a *trainable* frequency radius to dynamically separate and enhance complementary bands via three steps. First, for each spectral component indexed by i, we compute a normalized radial distance

$$\text{Dist}(i) = \frac{\sqrt{(X_i - X_c)^2 + (Y_i - Y_c)^2}}{\sqrt{X_{\max}^2 + Y_{\max}^2}} \tag{16}$$

where (X_c, Y_c) is the spectrum center and $(X_{\max}, Y_{\max})$ are the maximal coordinate indices along each axis, so $\text{Dist}(i) \in [0, 1]$ quantifies each component's position relative to the center. A trainable boundary radius r is mapped by a Sigmoid to $R = \sigma(r) \in [0, 1]$, partitioning the spectrum into LF/HF regions and preventing divergence during training. To maintain differentiability, we use a soft mask

$$\text{Mask}_{\text{hi}} = \sigma((\text{Dist} - R) \cdot \tau) \tag{17}$$

where $\tau = 10$ is a temperature hyperparameter. HF components are enhanced via a position-wise MLP

$$A_{\text{hi}} = \text{MLP}(A \odot \text{Mask}_{\text{hi}}) \tag{18}$$

and then, concatenated with the original amplitude A and fused by a 1×1 convolution with a residual update

$$A' = \text{Conv}_{1\times1}(\text{Concat}(A, A_{\text{hi}})) \quad A \leftarrow A + A' \tag{19}$$

where $A \in \mathbb{R}^{C\times H\times(\lfloor W/2\rfloor+1)}$ is the input amplitude spectrum, and MLP$(\cdot)$ is a channel-wise two-layer perceptron with nonlinearity. This preserves LF structure while adaptively reinforcing HF details carrying modality-complementary semantics. The learnable radius R self-adjusts to input characteristics, eliminating manual threshold tuning.

F. Loss Function

We integrate three objectives to jointly promote semantic accuracy, multiscale representation learning, and amplitude–phase coherence.

The primary segmentation loss is defined as the standard cross-entropy loss:

1) Primary Semantic Supervision:

$$\mathcal{L}_{\text{seg}} = -\sum_{i\in\Omega}\sum_{c=1}^{C} y_{i,c} \log \hat{p}_{i,c} \tag{20}$$

where Ω is the set of labeled pixels (excluding the ignore label 255), $y_{i,c} \in \{0, 1\}$ is the one-hot ground truth for class c, and $\hat{p}_{i,c}$ is the predicted probability.

2) Auxiliary Semantic Supervision: An auxiliary head at the penultimate stage ($t = 2$) uses the same objective

$$\mathcal{L}_{\text{aux}} = -\sum_{i\in\Omega}\sum_{c=1}^{C} y_{i,c} \log \hat{p}_{i,c}^{(a)} \tag{21}$$

where $\hat{p}_{i,c}^{(a)}$ denotes auxiliary predictions.

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TABLE III

QUANTITATIVE RESULTS (%) ON DDHR-SK. PER-CLASS IOU IS REPORTED FOR FIVE LAND-COVER CATEGORIES (BUILDINGS, ROADS, GREENERY, WATER, AND FARMLAND), ALONG WITH OA, MKAPPA, MF1, AND MIOU

	Method	Venue	Per-class IoU					OA	mKappa	mF1	mIoU
			Buildings	Roads	Greenery	Water	Farmland				
RGB	DANet [19]	CVPR'19	94.46	75.11	91.10	99.03	90.59	96.23	73.35	94.57	90.06
	CGNet [21]	TIP'20	80.19	27.57	72.77	90.47	69.97	86.16	33.90	78.76	68.19
	OCRNet [20]	ECCV'20	90.18	60.49	84.42	97.87	82.84	93.17	58.96	90.26	83.16
	SETR [23]	CVPR'21	94.17	74.10	91.43	99.18	90.88	96.22	73.79	94.49	89.95
	SegFormer [22]	NeurIPS'21	92.44	67.11	88.76	98.82	88.18	95.00	66.73	92.71	87.06
	Mask2Former [24]	CVPR'22	94.16	80.46	91.08	98.68	86.88	96.07	74.05	94.76	90.25
	PIDNet [25]	CVPR'23	87.77	59.01	80.57	96.89	77.64	91.44	53.43	88.56	80.38
	TransUNet [26]	MIA'24	90.67	62.68	85.64	98.41	83.71	93.64	61.82	90.95	84.22
SAR	DANet [19]	CVPR'19	91.20	58.97	85.93	98.13	82.31	93.57	59.35	90.28	83.31
	CGNet [21]	TIP'20	81.16	19.58	69.45	90.27	60.78	84.63	25.42	74.96	64.25
	OCRNet [20]	ECCV'20	86.38	38.70	78.36	96.18	71.34	89.52	41.07	83.54	74.19
	SETR [23]	CVPR'21	90.97	59.57	84.71	98.16	80.51	93.27	59.39	89.99	82.79
	SegFormer [22]	NeurIPS'21	88.81	46.92	81.99	97.57	76.75	91.51	49.55	86.73	78.41
	Mask2Former [24]	CVPR'22	91.34	58.52	86.40	93.29	80.08	93.20	62.69	89.49	81.92
	PIDNet [25]	CVPR'23	84.37	36.77	73.28	94.91	66.91	87.55	38.63	81.49	71.25
	TransUNet [26]	MIA'24	86.56	38.45	77.56	96.53	71.30	89.49	42.21	83.44	74.08
SAR & RGB	MCANet [29]	JAG'21	83.90	38.06	75.40	96.04	69.99	88.07	41.04	82.54	72.68
	DDHRNet [30]	JAG'22	82.76	29.45	73.78	95.20	67.40	87.21	35.24	79.81	69.72
	CEN [34]	TPAMI'22	86.61	40.08	79.34	97.57	74.26	90.20	46.91	84.50	75.57
	TokenFusion [55]	CVPR'22	92.28	65.60	88.05	98.60	85.74	94.63	64.71	92.10	86.05
	CMX [32]	TITS'23	93.50	72.03	90.02	99.00	88.03	95.59	70.48	93.65	88.52
	CDDFuse [13]	CVPR'23	82.19	35.87	69.95	93.94	54.50	85.06	66.26	78.55	67.29
	FTransUNet [6]	TGRS'24	86.78	40.51	78.70	97.24	73.13	89.91	46.02	84.35	75.27
	ASANet [33]	ISPRS'24	86.67	43.18	79.02	97.28	73.99	90.17	46.40	85.03	76.03
	MRFS [31]	CVPR'24	94.63	76.20	91.30	99.11	90.22	96.32	74.23	94.72	90.29
	ASMFNet [56]	JSTARS'24	86.24	37.52	79.08	97.45	73.38	89.92	45.24	83.77	74.74
	MIEFNet [57]	GRSL'24	90.56	63.90	85.20	98.39	82.19	93.46	61.20	90.89	84.05
	MFNet [58]	TGRS'25	89.79	56.65	83.61	98.10	80.32	92.67	57.54	89.23	81.70
	PAD (Ours)		**97.09**	**86.12**	**95.38**	**99.47**	**94.89**	**98.04**	**84.38**	**97.16**	**94.59**

3) PAD Constraint: To regularize the fused amplitude toward its decoupled counterpart across scales, we define

$$\mathcal{L}_{\text{amp}} = \frac{1}{|\mathcal{T}|} \sum_{t \in \mathcal{T}} \alpha_t \|\mathcal{A}'_t - \mathcal{A}_t\|_F^2 \tag{22}$$

where $\mathcal{T} = \{0, 1, 2, 3\}$ indices feature scales, $\alpha_t = 0.5 + 0.5\ t/3$ applies depth-progressive weighting, $\|\cdot\|_F$ is the Frobenius norm, and $\mathcal{A}_t$ and $\mathcal{A}'_t$ are the original and fused amplitude spectra at scale t, respectively.

The overall loss is formulated as follows:

4) Composite Objective:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{seg}} + \lambda_1 \mathcal{L}_{\text{aux}} + \lambda_2 \mathcal{L}_{\text{amp}} \tag{23}$$

where $\lambda_1 = 0.4$ balances auxiliary supervision and $\lambda_2 = 0.1$ controls amplitude regularization. This composite loss promotes geometric invariance by stabilizing phase-sensitive features via $\mathcal{L}_{\text{amp}}$, and semantic robustness via hierarchical amplitude refinement.

IV. EXPERIMENTAL RESULTS

A. Experimental Setup

1) Dataset Description: WHU-OPT-SAR [29] is a multimodal land-use dataset covering Hubei Province, China. It contains 100 coregistered RGB–SAR–NIR triples labeled with seven categories: farmland, urban, village, water, forest, roads, and others. RGB (GF-1) and SAR (GF-3) images are resampled to 5-m resolution and sized 5556×3704 pixels. Each image is cropped into nonoverlapping 512×512 patches, yielding 7000 samples split 80%/20% for training/testing.

DDHR-SK [30] is a fog-degraded subset of DDHR from Pohang, South Korea, pairing GF-2 RGB and GF-3 SAR images. It contains 6173 cloud-affected RGB–SAR pairs, resampled to 1 m and resized to 256×256 pixels, with five categories: buildings, farmland, greenery, water, and roads. The public split provides 3087 training and 3086 validation samples.

2) Training Settings: All models are trained using the MMSegmentation framework [54] on a single NVIDIA H800 GPU (comparisons use an NVIDIA A800). We use a batch size of 4 for 80k iterations, validating every 8k iterations. Optimization is performed with AdamW, an initial learning rate of 1×10^{-4}, and a cosine-annealing schedule (T_max = total iterations). A fixed seed (42) ensures reproducibility.

3) Evaluation Metrics: We report mean intersection-over-union (mIoU), overall accuracy (OA), $F1$ score, and Cohen's Kappa (mKappa), selecting the checkpoint with the highest mIoU. To mitigate class-imbalance bias, Kappa is computed for each test sample and then averaged across all samples to obtain mKappa.

B. Main Quantitative Results

To assess PAD's effectiveness, we replicate the original configurations of 20 comparison models, 12 state-of-the-art

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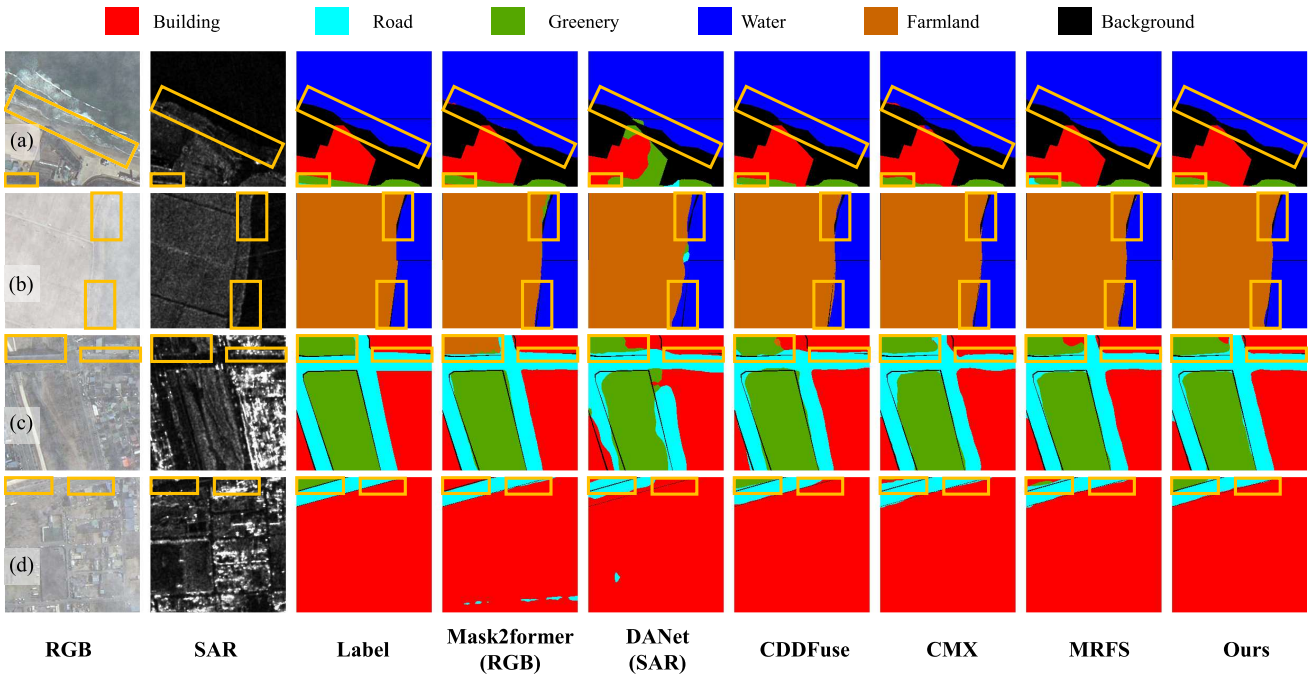


Fig. 7. Visualization results on DDHR-SK. Four boundary-focused scenarios (a) coastal demarcation, (b) water–cropland interface, (c) vegetation–building complex, and (d) road-network topology. For each scenario (left to right), the RGB input, SAR coherence map, pseudocolor ground truth, and predictions from six methods are shown. We compare PAD against unimodal baselines (Mask2Former-RGB and DANet-SAR) and multimodal approaches (CDDFuse, CMX, and MRFS). Critical boundary regions are highlighted with orange boxes; see Section IV-B for details.

multimodal segmentation approaches, and eight leading unimodal methods, on both WHU-OPT-SAR and DDHR-SK.

1) Evaluations on WHU-OPT-SAR: As shown in Table II, PAD achieves new state-of-the-art results: 56.26% mIoU, 84.56% OA, 53.93% mKappa, and 70.14% mF1. This surpasses the best unimodal method (DANet-RGB [19]: 53.25% mIoU and 83.19% OA) and the multimodal baseline CMX [32] (55.76% mIoU and 84.53% OA). For fine-grained classes, PAD attains 43.24% IoU on roads and 49.70% on villages, improving over MRFS by 1.48% and 0.66%, respectively.

Qualitative results in Fig. 6 further highlight PAD's benefits. In Fig. 6(a), PAD recovers occluded road segments (cyan) in building-dense areas via SAR phase cues, whereas RGB-only models miss them. In Fig. 6(b), PAD distinguishes specular reflections from true water boundaries under cloud interference. In Fig. 6(c) and (d), PAD resolves vegetation spectral ambiguities and maintains consistency across urban–rural transitions.

2) Evaluations on DDHR-SK: As shown in Table III, PAD exhibits superior performance under adverse (cloudy) conditions on DDHR-SK. It achieves 94.59% mIoU and 98.04% OA, outperforming leading unimodal methods (e.g., Mask2Former-RGB: 90.25% mIoU and 94.76% mF1, and DANet-SAR: 83.31% mIoU and 90.28% mF1) and the multimodal baseline MRFS (90.29% mIoU and 74.23% mKappa). PAD also attains 84.38% mKappa and 97.16% mF1. Boundary-sensitive categories achieve 86.12% IoU on roads, surpassing MRFS by 9.92%.

Visual results demonstrate PAD's ability for cross-modal boundary refinement. In Fig. 7(a), PAD accurately reconstructs periodically submerged coastlines via SAR coherence analysis, whereas RGB-only models confuse tidal sediments with terrestrial areas. In Fig. 7(b), PAD leverages SAR features to delineate water-infiltration boundaries in irrigated fields, overcoming the over-segmentation common in optical sensors. In Fig. 7(c) and (d), PAD produces continuous road networks under vegetation occlusion and precise material boundaries in vegetation–building complexes.

C. Ablation and Efficiency Analysis

1) Ablation Study: We conduct comprehensive ablations on WHU-OPT-SAR to validate the contributions of our core components: the ASF module and the PSC module.

a) Main Modules: Four variants (see Table IV) are evaluated. The baseline, which replaces substitutes PAD (ASF+PSC) with pixel-wise addition, achieves the lowest mIoU (55.37%). Enabling ASF alone reaches 56.11% mIoU; PSC alone provides 56.12% mIoU. Integrating two modules achieves the best performance (56.26% mIoU), demonstrating effective synergy.

b) Fusion Modules: To further validate ASF and PSC, we replace each with five existing frequency-domain designs, FeINFN [17], FCENet [15], FGSwin [59], Frefusion [16], and SFINet++ [50], while keeping the rest of our framework unchanged (see Table V). These methods represent strategies such as fixed-kernel spectral convolution (FeINFN), frequency–spatial interaction via Hadamard products/matrix multiplication (FCENet), learned spectral filtering (FGSwin), attention-based cross-domain fusion (Frefusion), and amplitude–phase parallel processing (SFINet++). Our

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TABLE IV

ABLATION RESULTS (%) ON MAIN MODULES FOR WHU-OPT-SAR. ✓ AND ✗ INDICATE THE INCLUSION OR EXCLUSION OF ASF AND PSC, RESPECTIVELY

Components		Per-class IoU							OA	mKappa	mF1	mIoU
ASF	PSC	Farmland	City	Village	Water	Forest	Road	Others				
✗	✗	69.46	57.16	49.01	65.15	83.87	41.57	21.39	84.32	51.96	69.23	55.37
✓	✗	69.91	**58.51**	49.53	65.52	84.05	42.63	22.58	84.55	**54.53**	69.93	56.11
✗	✓	69.98	57.32	49.36	**65.62**	**84.12**	**43.42**	23.01	**84.57**	53.80	69.98	56.12
✓	✓	69.85	58.09	**49.71**	65.39	84.05	43.24	**23.47**	84.56	53.96	**70.14**	**56.26**

TABLE V

ABLATION ON FUSION MODULES (%) FOR WHU-OPT-SAR. AMPLITUDE-ONLY, PHASE-ONLY, AND COMBINED (AMPLITUDE+PHASE) FUSION DESIGNS ARE COMPARED, EACH INTEGRATED INTO OUR FRAMEWORK

	Fusion module	Per-class IoU							OA	mKappa	mF1	mIoU
		Farmland	City	Village	Water	Forest	Road	Others				
Amplitude	FeINFN [17]	66.93	55.80	45.15	61.27	82.77	35.54	19.17	82.95	47.31	66.46	52.38
	FCENet [15]	68.27	55.18	46.79	61.17	83.43	37.09	16.71	83.43	49.82	66.52	52.66
	FGSwin [59]	69.87	57.43	49.03	65.04	**84.12**	**43.16**	22.26	84.51	53.53	69.70	55.85
	PAD (Ours)	**69.91**	**58.51**	**49.53**	**65.52**	84.05	42.63	**22.58**	**84.55**	**54.53**	**69.93**	**56.11**
Phase	FeINFN [17]	60.66	53.25	31.99	54.14	80.64	19.84	15.02	79.42	33.32	58.89	45.08
	Channel attention	69.58	**58.66**	**49.78**	64.28	83.96	**43.43**	22.65	84.39	53.54	69.93	56.05
	PAD (Ours)	**69.98**	57.32	49.36	**65.62**	**84.12**	43.42	**23.01**	**84.57**	**53.80**	**69.98**	**56.12**
Whole	FeINFN [17]	66.43	51.97	44.36	58.48	82.80	32.33	12.64	82.33	47.35	63.63	49.86
	FCENet [15]	66.87	53.01	46.41	60.30	82.64	30.21	14.51	82.78	43.83	64.33	50.56
	FGSwin [59]	69.88	57.86	49.48	65.26	**84.15**	42.49	22.36	**84.56**	53.41	69.76	55.92
	Frefusion [16]	**69.91**	57.99	49.35	64.22	84.00	42.72	22.46	84.41	**53.98**	69.69	55.81
	SFINet++ [50]	68.23	55.65	47.31	62.35	83.10	38.48	19.04	83.50	49.56	67.43	53.45
	PAD (Ours)	69.85	**58.09**	**49.71**	**65.39**	84.05	**43.24**	**23.47**	**84.56**	53.96	**70.14**	**56.26**

TABLE VI

ABLATION ON LOSS-FUNCTION WEIGHTS (%) FOR WHU-OPT-SAR. WE VARY λ_1 (WITH λ_2 FIXED AT 0.1) AND λ_2 (WITH λ_1 FIXED AT 0.4)

Configuration	Per-class IoU							OA	mKappa	mF1	mIoU
	Farmland	City	Village	Water	Forest	Road	Others				
Fixed $\lambda_2 = 0.1$											
$\lambda_1 = 0.01$	**69.87**	57.71	49.43	65.31	**84.12**	**43.35**	23.22	84.54	53.60	70.02	56.14
$\lambda_1 = 0.4$	69.85	**58.09**	**49.71**	**65.39**	84.05	43.24	**23.47**	**84.56**	**53.96**	**70.14**	**56.26**
$\lambda_1 = 1.0$	69.56	57.30	49.10	**65.39**	83.84	41.59	20.97	84.37	52.20	69.21	55.39
Fixed $\lambda_1 = 0.4$											
$\lambda_2 = 0.01$	69.83	56.80	48.67	**65.63**	**84.19**	43.17	22.82	84.52	53.54	69.76	55.87
$\lambda_2 = 0.1$	**69.85**	**58.09**	**49.71**	65.39	84.05	43.24	**23.47**	**84.56**	**53.96**	**70.14**	**56.26**
$\lambda_2 = 1.0$	69.83	57.74	49.37	64.95	83.95	**43.45**	22.50	84.46	53.87	69.84	55.97

ASF outperforms other amplitude-focused methods (56.11% versus 55.85% mIoU for FGSwin), and PSC consistently exceeds phase baselines (56.12% versus 56.05% for channel attention and 45.08% for FeINFN). At the full-fusion level, our integrated design still leads (56.26% mIoU), indicating that both modules are effective in isolation and synergistic when combined. These results confirm that adaptive frequency modulation (ASF) and spatially guided phase correction (PSC) offer structurally distinct and more effective alternatives to existing paradigms.

c) Loss Function Weights: We conduct weight ablations (see Table VI) to assess the trade-off between the amplitude consistency loss $\mathcal{L}_{\text{amp}}$ and the segmentation loss $\mathcal{L}_{\text{seg}}$. The best mIoU (56.26%) is achieved with $\lambda_1 = 0.4$ and $\lambda_2 = 0.1$. Increasing λ_1 above 0.4 degrades performance, especially on under-represented classes, whereas too small λ_2 significantly reduces accuracy, confirming that segmentation supervision must remain dominant.

d) Initial Frequency Radius: We vary the initial frequency radius r, which controls the preserved LF proportion in ASF (see Table VII). The optimal balance is at $r = 0.1$ (56.26% mIoU). Values that overly suppress or over-smooth LF content degrade performance, although the method remains robust for $r \in [0.1, 10]$.

2) Computational Efficiency: We evaluate computational efficiency in terms of model size (fusion modules only), inference time per image, and frames per second (FPS), summarized in Table VIII. We compare against three high-mIoU

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TABLE VII

ABLATION ON INITIAL FREQUENCY RADIUS (%) FOR WHU-OPT-SAR. THE TRAINABLE RADIUS PARAMETER IN THE ASF MODULE IS VARIED

Radius	Per-class IoU							OA	mKappa	mF1	mIoU
	Farmland	City	Village	Water	Forest	Road	Others				
0.01	**70.08**	57.44	49.31	**66.29**	**84.18**	43.08	22.25	**84.66**	53.92	69.88	56.09
0.1	69.85	58.09	**49.71**	65.39	84.05	43.24	23.47	84.56	**53.96**	**70.14**	**56.26**
1	69.79	**58.19**	**49.71**	64.81	84.15	**43.44**	23.03	84.52	53.88	70.04	56.16
10	69.71	57.71	49.69	65.63	**84.18**	43.21	**23.56**	84.57	53.93	70.12	56.24
100	69.90	57.23	49.57	65.23	84.13	43.34	22.94	84.53	53.88	69.93	56.05

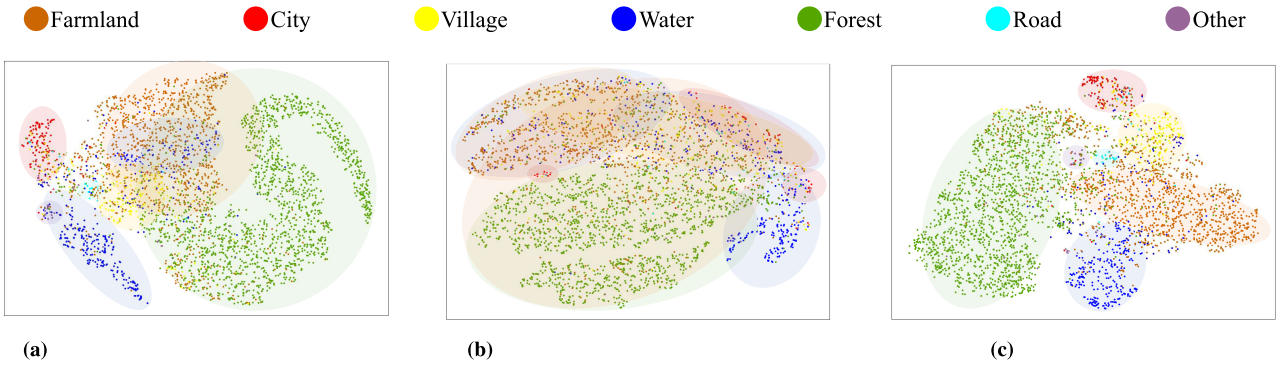


Fig. 8. t-SNE visualization of cross-modal feature distribution evolution. (a) RGB-encoder features show spectral continuity but exhibit blurred class boundaries, (b) SAR-encoder features capture geometric details but are affected by speckle noise, and (c) PAD-fused features demonstrate enhanced intraclass compactness and improved interclass separability.

TABLE VIII

COMPUTATIONAL EFFICIENCY COMPARISON. MODEL SIZE (#PARAMS), PER-IMAGE INFERENCE TIME, AND THROUGHPUT (FPS) FOR COMPETING METHODS. FEWER PARAMETERS AND SHORTER INFERENCE TIMES, ALONG WITH HIGHER FPS, INDICATE BETTER EFFICIENCY

Method	#Params (M)	Inference time (ms)	Throughput (FPS)
CMX [32]	15.00	206.70	1,957.8
MRFS [31]	10.30	695.85	1,342.2
ASANet [33]	7.82	72.91	2,245.2
PAD (Ours)	**3.88**	**5.16**	**3,228.7**

methods, CMX [32], MRFS [31], and ASANet [33]. All metrics are aggregated across fusion stages with the same baseline model.

As shown in Table VIII, PAD uses only 3.882 M parameters, achieves the fastest inference time (5.16 ms per image), and delivers the highest FPS (3228.7). These results indicate that PAD not only matches or exceeds the accuracy of existing methods but also offers real-time efficiency suitable for operational deployment.

D. Further Analysis and Visualization

1) Visualization at the Distribution Level: To further analyze feature distributions, we extract multiscale features from the RGB and SAR encoders and from the fusion modules. After upsampling to the original resolution, we concatenate the features channel-wise. To balance computational efficiency with discriminative power, we apply a two-stage dimensionality reduction: 1) random spatial sampling and 2) principal component analysis (PCA), retaining 50 components (over 89% variance). The final visualization is generated using t-SNE [60] (perplexity = 30 and 1000 iterations).

As shown in Fig. 8, PAD-fused features exhibit enhanced intraclass compactness and improved interclass separability. Quantitatively, mean intraclass standard deviation is reduced by 20.39% for roads, 13.16% for water, 6.46% for forest, and 4.43% for farmland compared with the best unimodal baselines. Moreover, Euclidean distances between critical class pairs increase significantly: city/village by 75.75% (66.56 → 116.95) and farmland/forest by 13.29% (107.64 → 121.89), indicating that PAD captures cross-modal discriminative patterns effectively.

2) Robustness of PSC Under Challenging Scenarios: To further evaluate the robustness of PSC beyond normal distributions, we conducted targeted ablation experiments under extreme long-tail conditions, including missing RGB regions and cloud obscuration. Results are summarized in Table IX.

a) Missing Regions: When large portions of the RGB modality are absent, models without PSC accumulate residual phase errors, particularly in categories with fine spatial structures (e.g., forest and road). With PSC, alignment remains stable and yields consistent improvements across macro-level metrics (e.g., + 0.19 mF1 and + 0.25 mIoU).

b) Cloud Obscuration: Under cloud occlusion, traditional coregistration often provides weak or misleading guidance. PSC mitigates this issue by preserving phase consistency, resulting in systematic improvements in category-wise IoU (e.g., city and village) as well as overall agreement scores.

These results confirm that PSC not only aligns with physical and statistical expectations but also maintains robust

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TABLE IX
PERFORMANCE COMPARISON (%) OF MODELS W/ AND W/O PSC UNDER MISSING RGB REGIONS AND CLOUD OBSCURATION ON WHU-OPT-SAR

Scenario	Method	Per-class IoU							OA	Kappa	mF1	mIoU
		Farmland	City	Village	Water	Forest	Road	Others				
Missing RGB	*w/o* PSC	**74.65**	75.91	58.03	80.18	56.84	57.81	59.49	82.28	70.94	79.23	66.13
	w/ PSC	**74.65**	**76.25**	**58.05**	**80.25**	**57.09**	**58.54**	**59.82**	**82.35**	**71.12**	**79.42**	**66.38**
Cloud Obscuration	*w/o* PSC	75.33	75.47	57.42	76.31	84.30	53.40	44.94	86.90	63.06	79.23	66.74
	w/ PSC	**75.37**	**75.73**	**57.56**	**76.47**	**84.32**	**53.68**	**45.30**	**86.95**	**63.52**	**79.37**	**66.92**

TABLE X
STATISTICAL COMPARISON BETWEEN MODELS *w/* AND *w/o* ASF

Metric	*w/* ASF	*w/o* ASF
Mean Pearson correlation	0.0364	0.0369
Mean Spearman correlation	0.0482	0.0488
Paired t-test p-value	0.0016	
Wilcoxon test p-value	6.4×10^{-7}	
Cohen's d (effect size)	-0.038	

performance under adverse long-tail scenarios, where purely physical models typically degrade.

3) Noise Sensitivity of ASF: To assess whether ASF amplifies HF speckle noise in SAR images, we perform a correlation analysis using the equivalent number of looks (ENL), a standard measure of speckle. For each image, we compute pixel-wise ENL maps and then measure Pearson and Spearman correlations between ENL and classification accuracy. As shown in Table X, the model with ASF exhibits nearly identical correlation coefficients to the baseline; differences in mean correlation are below 0.001, and Cohen's $d = -0.038$ indicates a negligible effect size. These results confirm that ASF does not increase noise sensitivity and remains robust in low-SNR regions.

Overall, these findings confirm that PAD significantly improves classification accuracy and prediction consistency by integrating multimodal information, suppressing noise, and preserving discriminative spectral and geometric characteristics inherent in the data.

V. CONCLUSION

In this article, we proposed the PAD framework for fusing SAR and RGB modalities in LCC. Our frequency-domain analysis quantitatively demonstrates that phase components remain consistent across modalities, whereas amplitude spectra capture modality-complementary details, providing the first physically grounded prior for multisource remote-sensing fusion. PAD decouples and aligns phase features via the PSC module and adaptively fuses amplitude components using the ASF module, effectively leveraging SAR's HF details together with RGB's LF context. Extensive experiments on two challenging datasets show that PAD outperforms existing methods in both accuracy and robustness.

Despite its demonstrated effectiveness, PAD assumes well-aligned input modalities and relies on hard-label supervision. In future work, we will integrate self-supervised cross-modal registration refinement and soft-label modeling, and investigate transformer-based frequency fusion to further enhance global context awareness.

ACKNOWLEDGMENT

Numerical calculations were partly carried out on the Big Earth Data Cloud Service Platform.

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Huiling Zheng received the B.S. degree in marine science from Shanghai Ocean University, Shanghai, China, in 2019. She is currently pursuing the M.S. degree in computer science with the School of Computer Science and Artificial Intelligence, Wuhan University of Technology, Wuhan, China.

Her research interests include remote sensing image processing and AI oceanography.



Xian Zhong (Senior Member, IEEE) received the B.S. degree in computer science from Wuhan University, Wuhan, China, in 2007, and the Ph.D. degree in computer science from Huazhong University of Science and Technology, Wuhan, in 2013.

He was a Visiting Scholar at Peking University, Beijing, China, from 2021 to 2022, and at ROSE@EEE, Nanyang Technological University (NTU), Singapore, 2024. He is a Professor with the School of Computer Science and Artificial Intelligence, Wuhan University of Technology, Wuhan. His research interests include multimodal information processing, neuromorphic vision, and intelligent transportation systems.

Dr. Zhong serves as an Associate Editor for *TVC* and an Area Chair for IJCAI, ICASSP, ICME, ICIP, and IJCNN.



Bin Liu (Member, IEEE) received the B.S. degree in information engineering and the M.S. and Ph.D. degrees in signal and information processing from Shanghai Jiao Tong University, Shanghai, China, in 2007, 2009, and 2015, respectively.

He was a Visiting Ph.D. Student with the Department of Signal and Image Processing, Télécom ParisTech, Paris, France, from 2012 to 2013. He was an Assistant Researcher with Shanghai Key Laboratory of Intelligent Sensing and Recognition, Shanghai Jiao Tong University, from 2015 to 2018, and a Post-Doctoral Fellow with the College of Marine Sciences, Shanghai Ocean University, from 2018 to 2020, where he is currently an Associate Professor and serves as the Deputy Director of the Department of Marine Technology. His research interests include AI oceanography, SAR and PolSAR image processing and understanding, image segmentation/classification, and AI-empowered fishery forecasting.



Yi Xiao (Member, IEEE) received the B.S. degree from the School of Mathematics and Physics, China University of Geosciences, Wuhan, China, in 2020, and the M.S. and Ph.D. degrees from Wuhan University, Wuhan, China, in 2022 and 2025, respectively.

He is currently with the School of Computer and Artificial Intelligence, Zhengzhou University, Zhengzhou, China. His research interests include remote sensing image/video processing and computer vision. More details are available at https://xy-boy.github.io



Bihan Wen (Senior Member, IEEE) received the B.Eng. degree from Nanyang Technological University (NTU), Singapore, in 2012, and the M.S. and Ph.D. degrees from the University of Illinois at Urbana-Champaign (UIUC), Champaign, IL, USA, in 2015 and 2018, respectively.

He is a Nanyang Assistant Professor with the School of Electrical and Electronic Engineering, NTU. His interests include machine learning, computational imaging, computer vision, image processing, and AI security.

Dr. Wen's honors include the NTU Early Career Teaching Excellence Award 2022, the NTU Inspirational Mentor for Koh Boon Hwee Award 2021, the PEB Gold Medal 2012, the Best Paper Runner-Up at IEEE ICME 2020, the Best Paper at IEEE ICIEA 2023, and the Best Paper at IEEE MIPR 2023. He has been listed among Stanford University's World's Top 2% Scientists (AI) in 2021 and 2023 and was the 2023 CASS VSPC Rising Star Runner-Up. He is an Associate Editor for TCSVT and a Guest Editor for SPM and JSTSP.



Xiaofeng Li (Fellow, IEEE) received the B.S. degree in optical engineering from Zhejiang University, Hangzhou, China, in 1985, the M.S. degree in physical oceanography from the First Institute of Oceanography, Qingdao, China, in 1992, and the Ph.D. degree in physical oceanography from North Carolina State University, Raleigh, NC, USA, in 1997.

From 1997 to 2019, he was with the National Oceanic and Atmospheric Administration (NOAA), Washington, DC, USA, working on operational satellite ocean remote sensing products. He is currently with the Institute of Oceanology, Chinese Academy of Sciences, Qingdao, China. His interests include satellite/AI oceanography, big data, and image processing.

He is a member of the Editorial Board of IJDE, *Big Earth Data*, and JOL. He is also serves as an Associate Editor for TGRS and IJRS and the Executive Editor-in-Chief for the *Journal of Remote Sensing*.

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